

MCP User Manual

1 Introduction

Syntermed's Master Control Program (MCP) is a Windows Desktop application that serves as the central hub for managing licensing, applications, data and workflow. Once installed, MCP will assist the user in the following tasks:

- Licensing
- Application Updates
- Patient/Study Management
- Syntermed Live Sync
- Advanced Workflow

2 Installation

If your system is online, the installation process for Syntermed's latest suite of products is fast and easy. Please follow the instructions in this section to download, install and license our application(s).

Our latest applications are capable of running on older hardware, but have been designed to take full advantage of modern multi-processor / multi-core CPUs. Please refer to the [System Requirements](#) section of our website for more information.

2.1 Download / Install

Please download the latest version of MCP [here](#). The file will be in compressed, zip format.

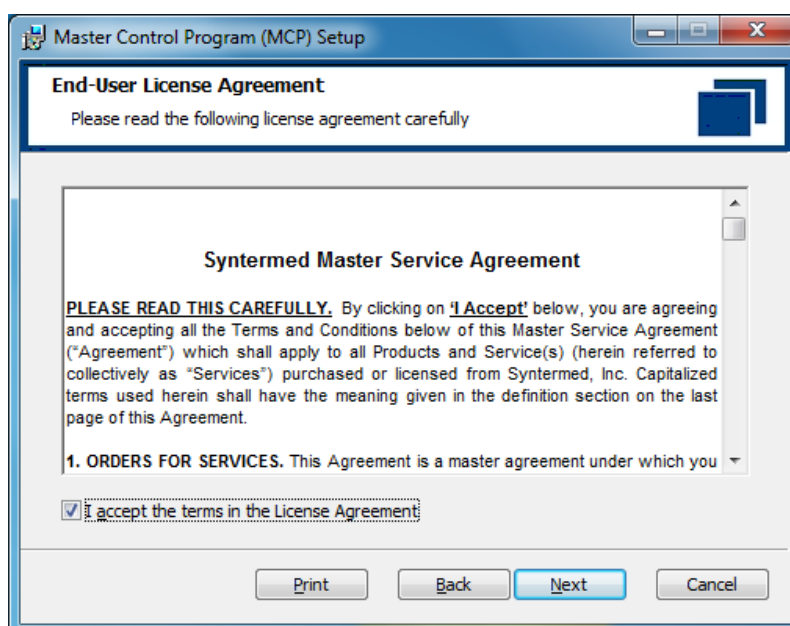
This is the latest, publicly available version. In some instances, you may have been provided with a different version. If the is the case, please follow the rest of these instructions using that installation package.

Once the download has completed, extract the file anywhere on your system, then run **setup.exe**.

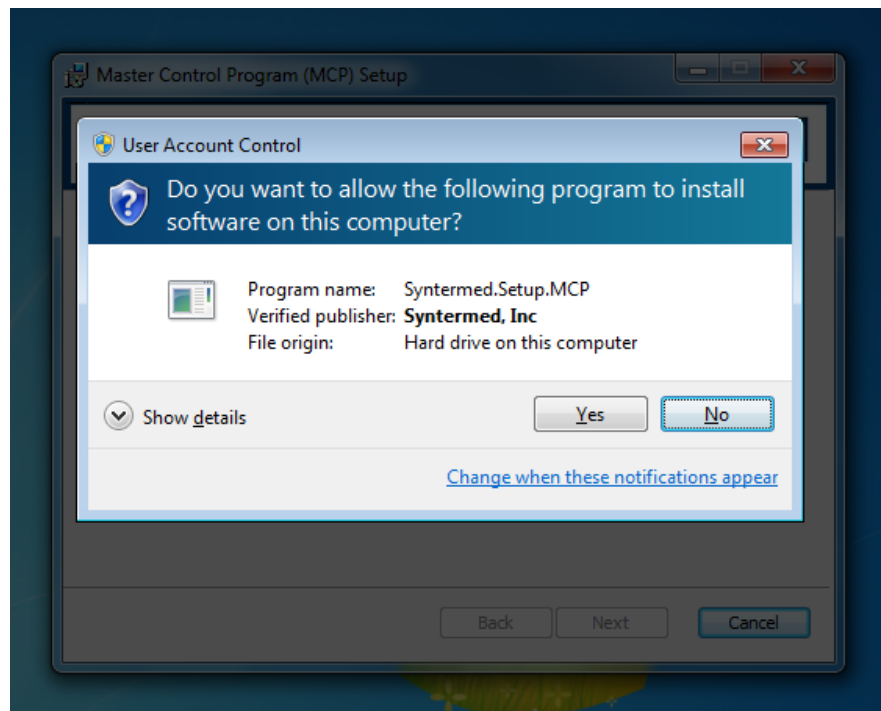
Name	Type	Compressed size
 setup	Application	188 KB
 Syntermed.Setup.MCP	Windows Installer Package	26,066 KB

Caution: The installation package will automatically detect any missing pre-requisites and download and install them for you. This includes the Microsoft .net Framework. An active Internet connection and possible reboot will be required for this to complete successfully, but will only be required on systems running Windows XP.

Next, the installation wizard will guide you through the process. You will be required to agree to the **Syntermed Master Services Agreement (MSA)** for the installation to complete successfully.

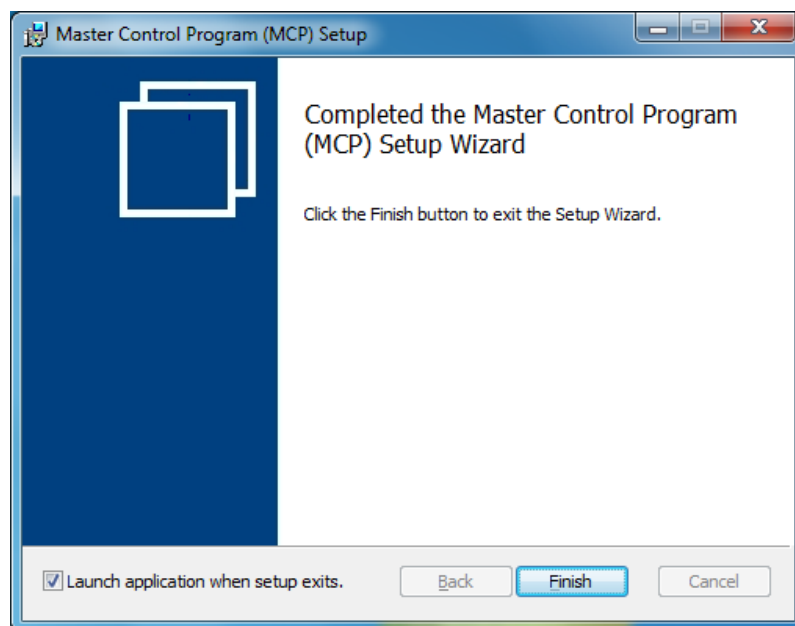


Depending on your system's configuration (please ask your IT department for more information) you may see the following dialog during installation. This is an expected additional security step built into the new Windows systems related to User Access Control (UAC). Please click **Yes** to continue.

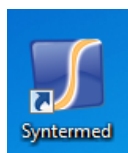


Caution: Please pay close attention to your computer during the installation process. If the UAC dialog (above) is left idle for a certain amount of time, it will close automatically and the installation will fail without providing the user with any notifications. This is a limitation within Microsoft Windows.

When the installation process has completed, the wizard will launch MCP for you automatically. This can be disabled by unchecking the **Launch application when setup exits** check box in the dialog below.



A Syntermed icon will also be placed on the Desktop. You can launch MCP by double-clicking on this icon.



2.2 Activate License

Upon initial installation, MCP will run in **Unlicensed** mode. The status of your License is displayed in the lower left status bar as defined below.



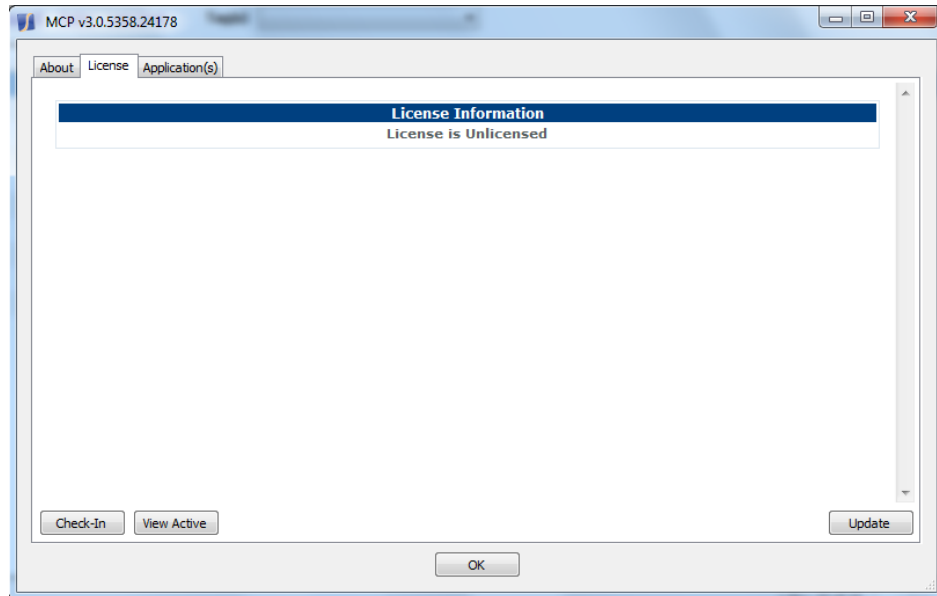
Unlicensed



Valid License

Use the **Licensing Module** within MCP to activate your license.

- Select the **About...** item from the **Help** menu in MCP to bring up the About dialog.
- Click on the **License** tab to view current license info (if applicable).
- Click the **Update** button in the lower right corner of the dialog to bring up the **Licensing Module**.

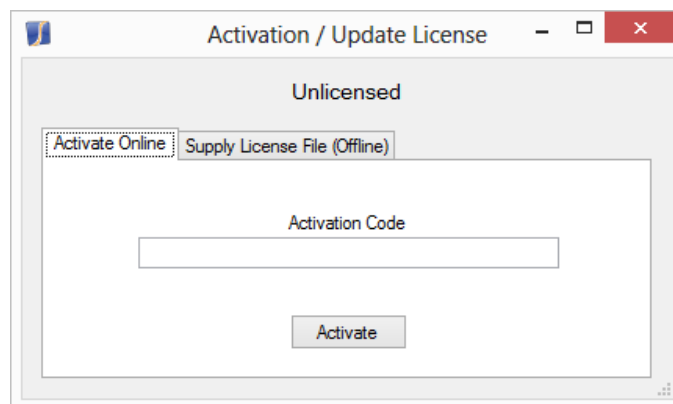


TIP: Clicking on the License Status icon in the status bar, will automatically open the License dialog.

Online Activation

Most commonly, Syntermed will provide an **Activation Code** which will allow MCP to retrieve the licensed components automatically. This requires that your computer is online.

To Activate your license, and enter your **Activation Code** on the **Online Activation** tab of the Licensing Module and click the **Activate** button. Please be patient as it may take a few seconds to communicate with our servers during this step.

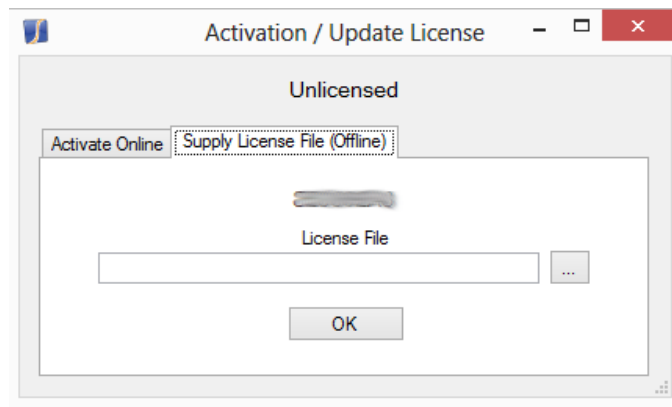


Offline License File

If your computer cannot access the Internet, Syntermed can provide physical license file, through the following two step process.

First, you must provide Syntermed with your **Secure Code** which can be found on the **Supply License File (Offline)** tab of the **Licensing Module**. Once this information has been provided, a license file (.lic) will be generated and returned.

Use the "..." button on the **Supply License File (Offline)** tab of the **Licensing Module** to navigate to the license file, then click OK.



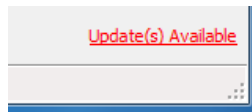
Summary

Once the licensing process is complete, the **Licensing Module** will automatically close and the up-to-date information about the current License will be displayed in the License dialog.

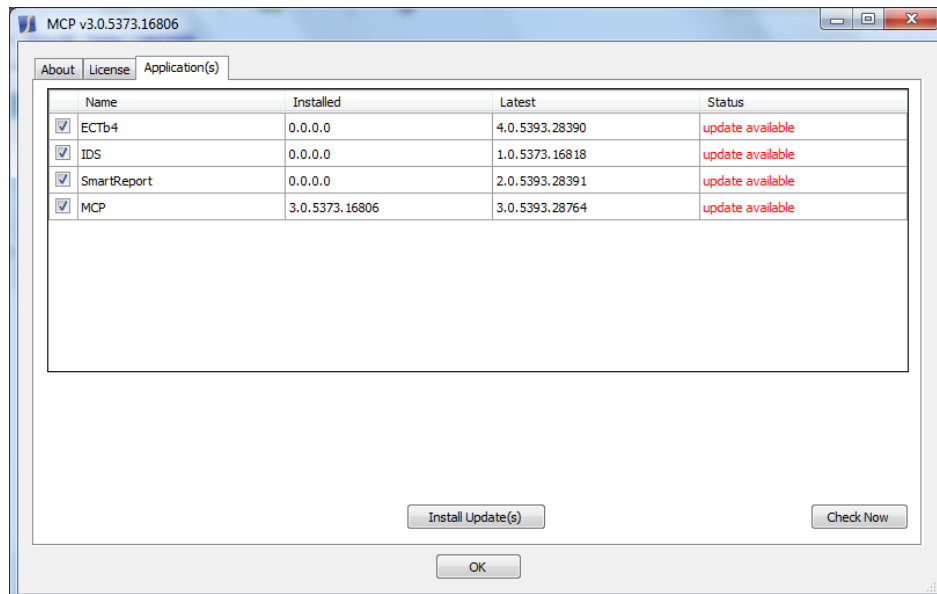
After the licensing process has been successfully completed, MCP must be re-started.

2.3 Update / Upgrade

By default, MCP will check for application updates at startup. Therefore, in the case of a newly activated license, as in the previous step, the **Update(s) Available** notice should be visible in the bottom right corner of the MCP window.

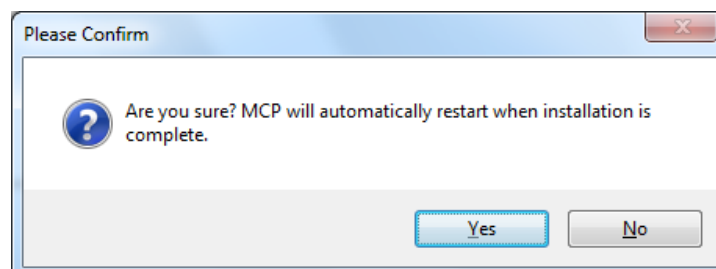


Clicking **Update(s) Available** will take you to the **Application(s)** dialog, where you will see what application(s) are available as well as the related version information and update status.



By default all applications with updates available will be selected via the check box on the far left of each row. You may toggle each check box to include or exclude updates. Click the **Install Update(s)** button to initiate the download and update procedure.

MCP must restart to complete the update process, which you will be asked to confirm.



The status of each download and update will be shown in real time on the Application(s) dialog.

Please note that only MCP has an installation wizard that requires user input. If you are not installing MCP, all application(s) will install quietly and MCP will restart automatically when the process is complete.

Caution: Depending on your security settings, you may see the UAC confirmation prompts as described in the **Download /Install** section.

2.4 Repair & Cleanup

Remove License File

If there is an unresolvable problem with the license, it may be necessary to clear the existing license file and start the licensing procedure over. The license file is located in the following path:

Windows Vista or Higher

C:\ProgramData\Syntermed\MCP\license.lic

Windows XP

C:\Documents and Settings\All Users\Application Data\Syntermed\MCP\license.lic

Clear the Database and/or Image Data Store

All DICOM images and/or physical files are automatically moved to a local Data Store when imported. All database entries are stored in a single .sdf file. All of this data can be cleared without effecting the license or settings by deleting the following file and folder:

Windows Vista or Higher

C:\ProgramData\Syntermed\MCP\Syntermed.Clinical.Desktop.sdf

C:\ProgramData\Syntermed\MCP\Data\

Windows XP

C:\Documents and Settings\All Users\Application Data\Syntermed\MCP\Syntermed.Clinical.Desktop.sdf

C:\Documents and Settings\All Users\Application Data\Syntermed\MCP\Data

Clear Everything

In the event that an installation has become corrupted, it may be necessary to uninstall and clear all data and resources related to Syntermed.

First, uninstall all Syntermed applications via the "Programs" utility in the Windows Control Panel. After uninstallation is complete, manually delete the following directories (if they exist).

Windows Vista or Higher

C:\ProgramData\Syntermed\MCP\

C:\Program Files\Syntermed\MCPv3\

C:\Program Files (x86)\Syntermed\MCPv3\

Windows XP

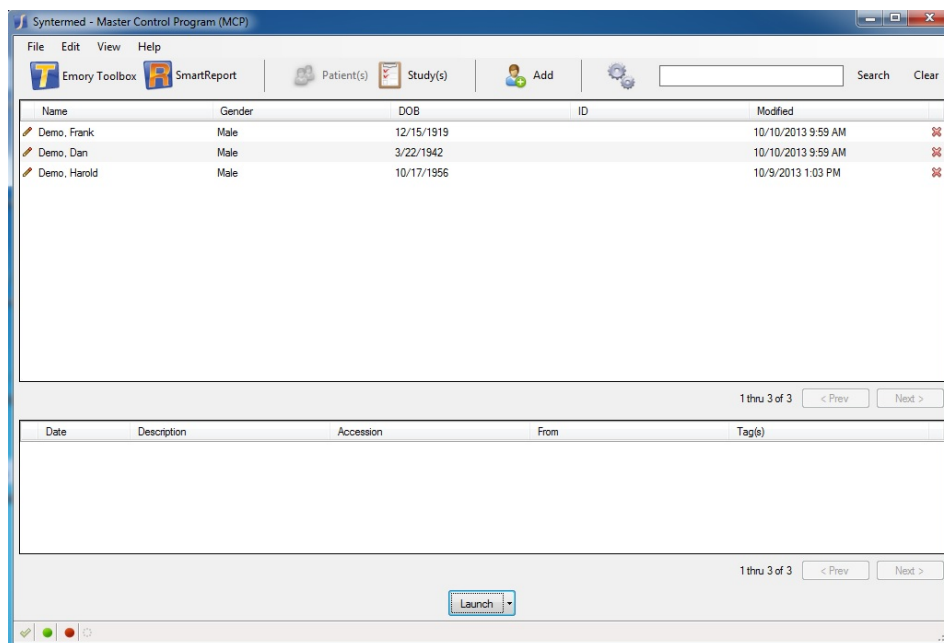
C:\Documents and Settings\All Users\Application Data\Syntermed\MCP\

C:\Program Files\Syntermed\MCPv3\

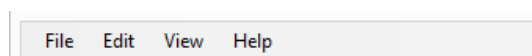
C:\Program Files (x86)\Syntermed\MCPv3\

3 The Display

The Master Control Program (MCP) provides a method to manage, display, and launch the patient data into applications and the MCP display is shown below.



The **Main Menu** bar provides a standard interface to access options via File, Edit, View, and Help.

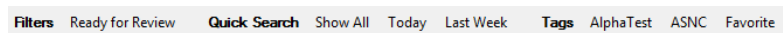


The **Toolbar** contains buttons for the most common actions, including:

- Quickly launching selected data in plugin/application (e.g. Emory Toolbox, SmartReport)
- Swap between Display views (by Patient or by Study)
- Add patient(s) to the database (Add button)
- Options dialog (Gears button)
- Search the database by keyword, date or date range



The **Filters, Quick Search and Tag(s)** bar is only available from within the **Study** view.



The main **Display** area contains a list of the patients and/or studies depending on the selected view. Please refer to the next section for more information.

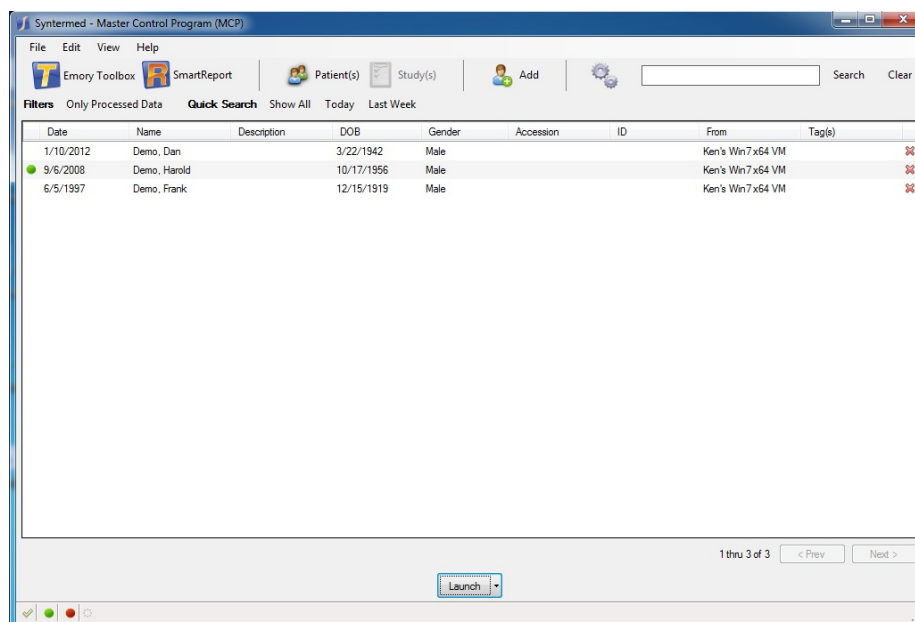
The bottom **Status** Bar consists of a few icons designed to give the user access to status of different parts of the application. From left to right:

- License status: Check mark = everything is OK, Caution symbol = there is a problem
- Connectivity status: Green dot = everything is OK, Red dot = the system is offline or the Syntermed Live server(s) cannot be reached
- DICOM server status: Red dot = DICOM server is stopped, Green dot = DICOM server is running
- Sync status: Red spinner = metadata is being synced with Syntermed Live, Blue spinner = images are being synced with Syntermed Live, Gray dormant spinner = no sync in progress



3.1 Study View

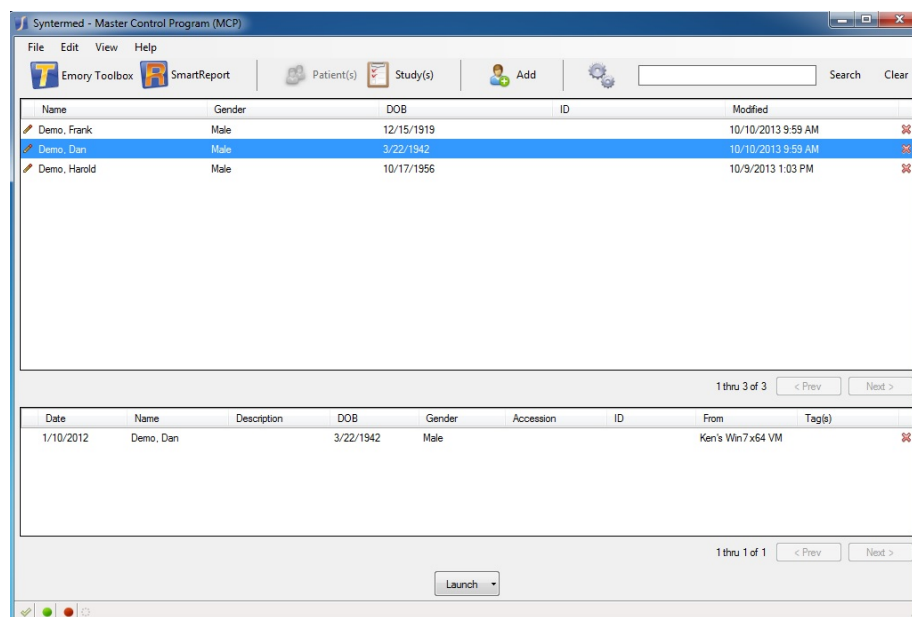
In the **Study** view, the user is presented with a list of studies with the most important properties of each study broken up into columns on which the data can be sorted. By default, the Study Date, Patient Name, DOB, Gender, and ID (primary) are displayed. There are many more columns available, such as Description, Tag(s) and where the study originated From (Synterm Live only) just to name a few. Columns can be added and removed from this view by right-clicking the header cell of any column and checking/unchecking the desired column from the menu. MCP will automatically remember your choices the next time you load the program.



Studies can be interacted with in a variety of ways. For example, a study can be launched into a plugin such as Emory Toolbox or SmartReport. Its properties can be modified by choosing "Study Details..." from the context menu which is displayed when the item is right-clicked. Some actions have multiple methods in which they can be performed, for instance a study can be deleted by either clicking the "X" icon on the far right of the row, or by selecting the "Delete" menu item from the context menu. The application is designed to allow the user to choose the most efficient way to perform the necessary task.

3.2 Patient View

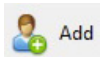
The patient view is shown below.



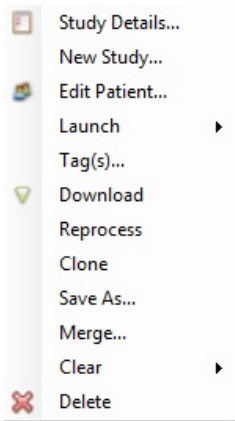
4 Working with Data

4.1 Add / Edit

Add Patient - allows for manually entering a patient into the database. To launch click on the Add Icon shown below. See section 4.1.1 for more information.



The edit patient and edit study section can be accessed by right clicking on the patient in the list and the following dialog box will be displayed.



Edit Patient - right click on the patient you wish to edit and then select Edit Patient from the list. See section 4.1.1 for details on editing the patient.

Edit Study - right click on the patient you wish to edit and then select Study Details from the list. See section 4.1.2 for details on editing the patients study.

4.1.1 Patient Info

The Add/Edit patient screen is shown below. If this is a new patient then you can add in the pertinent demographic information for the patient on this screen. When you are done click on the Save button at the bottom to save the patient in the database. If you wish to edit an existing patient then right click on the patient and select Edit Patient. This screen will come up with the current patient information filled out on the form which can then be edited.

A screenshot of the 'Add/Edit Patient' dialog box. It has three tabs: 'General' (selected), 'Advanced', and 'Import'. The 'General' tab contains the following fields: First Name, Last Name, ID, Phone, Email, Address, DOB, Gender (dropdown), Race, ID (secondary), Phone (secondary), Email (secondary), and Address (secondary). There is a 'Save' button at the bottom.

Advanced edit options is used to edit an existing patients study.

Add/Edit Patient

Advanced

Import

Name

DOB

Gender

Race

Identifier(s)

New...

Address(es)

New...

Phone Number(s)

New...

Email Address(es)

New...

Save

4.1.2 Study Details

Study Details - this screen displays the details for the particular study and these items can be edited by clicking on the pencil icon or by typing in the text boxes.

Plugin Data - This list displays the plugin associated with the study and the version number.

Files - the particular files associated with the study are listed in this box.

The 'Edit Study' window is divided into several sections:

- Study Info:** Fields for Patient, Study Date (9/6/2008), Tag(s), Status (Ready for Review), and Description.
- StudyInstanceUID:** 1.3.6.1.4.1.25403.220314657329294.13604.20130924125719.1
- AccessionNumber:** Ken's Win7 x64 VM
- Additional Info:** A table with columns Type, Name, and Value. It lists Height (65) and Weight (155).
- Plugin Data:** A table listing plugins: LvxStudy (IDS, 0.1.5024.20152), LvxNpStudyInfo (IDS, 0.1.5024.20152), NpStudyInfo (SmartReport, 2.0.5031.27465), EcStudy (Emory Toolbox, 4.0.5031.27464), and NpReport (SmartReport, 2.0.5031.27465).
- File(s):** A list of files with columns Type and Name. It includes DCM files for Rest PlanarProjection Gated, Rest ShortAxis Gated, Rest ShortAxis, Rest ShortAxis AC, Rest Transmission, Stress PlanarProjection Gated, Stress ShortAxis Gated, Stress ShortAxis, Stress ShortAxis AC, and Stress Transmission.

4.2 Importing

Importing pertains to how patient data can be imported into the MCP application. There are multiple ways to do this and they are explained in this section.

4.2.1 DICOM Server

The DICOM window can be accessed by going to Edit ->Options->DICOM. Configure the options in the DICOM Server sect

The 'Options' window has tabs for General, DICOM, External Application(s), and Proxy. The 'DICOM' tab is active, showing the following settings:

- DICOM Parsing:** A list of image types (Image Type, Acquisition, Orientation, Isotope, Radiopharmaceutical, AC, MC, Dynamic) with a 'Reset' button.
- DICOM Server:** A section with checkboxes for 'Enable DICOM Server', 'AE Title' (MCP), and 'Port' (104).
- Patient Name Parsing:** A section with checkboxes for 'Enable DICOM Patient Name Parsing', 'Incoming format = Last Name appears first', and 'Display Format = Last Name'.

Steps for setting up and transferring DICOM data into MCP.

1. Set up a DICOM node on the nuclear OEM workstation. All of the nuclear OEM's provide a method for setting up a DICOM node on their workstations so that patient data can be converted to DICOM and pushed to another system with a DICOM server. In order to do this there are 3 variables required to allow the two system to interact with each other. They are the AETitle, port #, and IP address. The AETitle is MCP, port # is 104 and the IP address is the IP address of the PC where MCP is installed. Set up the DICOM node on the OEM workstation by entering these 3 values into the node that is being set up.
2. On the PC where MCP is installed go to the DICOM Server section and make sure that the AE Title and Port number are the same as the ones used on the OEM workstation. Also check the Enable DICOM Server box so that the server will start up whenever MCP is started.
3. On the OEM workstation push the patient file over to the DICOM node called MCP and the study will show up in the MCP window. If this box is not checked then you will have to start/stop the DICOM server using the icon (green dot / red dot) in the bottom status bar of MCP.

The DICOM server is stopped here shown as the red dot, to start the server click on the red dot and it will turn green indicating the DICOM server is running.

Note: At least one **Watch Folder** must be configured for the DICOM Server to function properly. Please see the next section for details on how to create a **Watch Folder**.

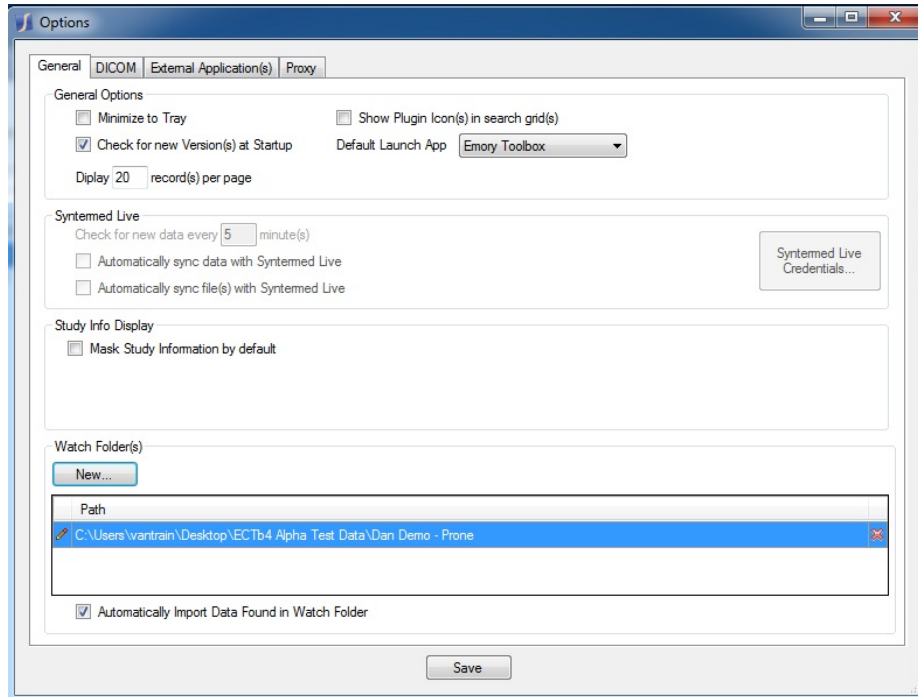
4.2.2 Watch Folder(s)

Configure watch folders in the MCP Options dialog.

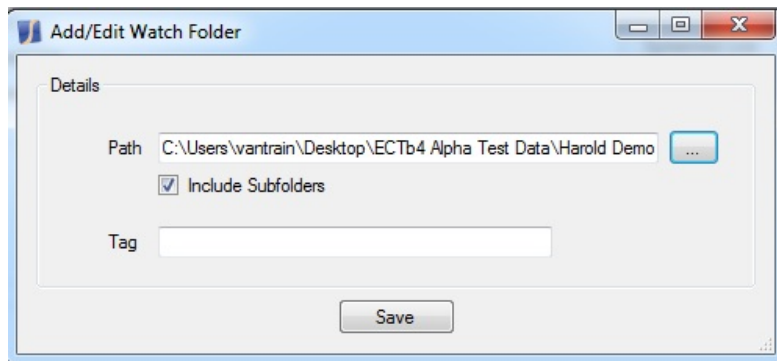
You can have as many watch folders as you would like.

If MCP is running, as soon as data arrives in a watch folder, it will automatically be imported. If MCP is not running, all watch folders will be analyzed when the application starts.

Each watch folder can have associated **Tags**.



New Watch Folder - to create a new watch folder click on the New button and the screen shown below will be displayed. Either type in the path or click on the  button and then browse to the path and select to enter it into the Path box. If you wish to enter a Tag for the watch folder then this can be entered in the Tag box. When you are done click on the Save button to save the watch folder path.



4.2.3 Drag & Drop

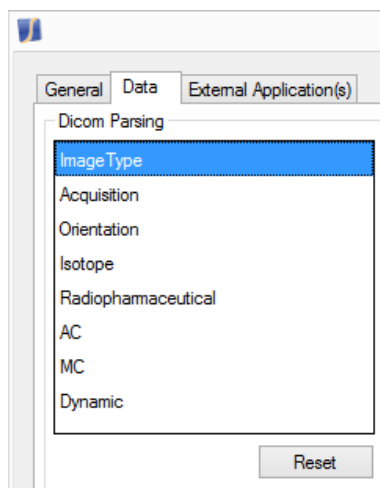
Any study can also be imported into the MCP database by selecting the patient study, dragging and then dropping it into the MCP window. The patient study will then be imported and displayed in the MCP window.

4.3 DICOM Parsing

The application attempts to automatically determine the following attributes of each DICOM file based on the configurable rules described in this section.

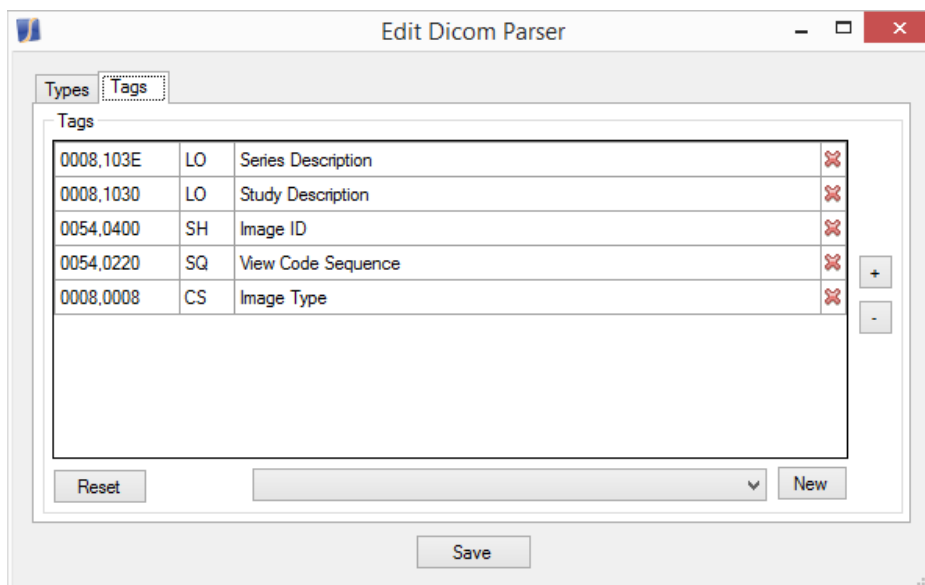
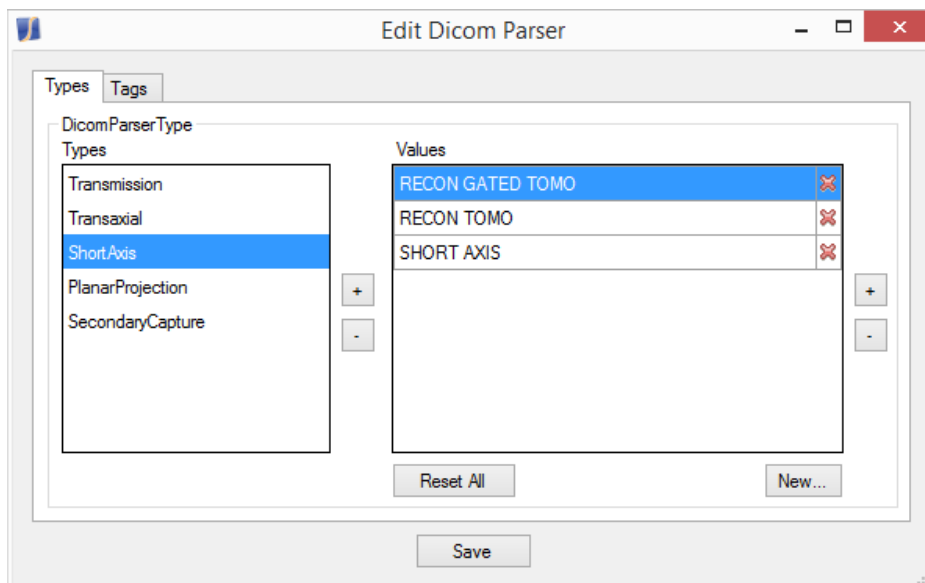
- Image Type
- Acquisition Type
- Orientation
- Isotope
- Radiopharmaceutical
- AC (Attenuation Corrected)
- MC (Motion Corrected)
- Dynamic

These rules can be modified by navigating to be the **Options** dialog in MCP, then clicking on the **Data** tab. The DICOM Parsing section is shown below.



TIP: If you are consistently modifying one or more of these properties on the **Verify** screen in the **Emory Toolbox** application, you should be able to configure the application to properly auto detect these properties based on your specific data based on the information in this section.

Each property has a list of **Types** that are defined by looking for keyword(s) in specific tags in the file's DICOM header. To bring up the configuration dialog for a property type, simply double-click on the item. In the configuration dialog for that property, click on the Type on the left to see a list of the keyword(s) (or **Values**) being searched. Click on the **Tags** tab to see the list of tags being searched.



In the above example, you can see that when detecting the "Short Axis" Image Type, the Series Description, Study Description, Image ID, View Code Sequence and Image Type tags are being searched for the keywords "RECON GATED TOMO", "RECON TOMO" and "SHORT AXIS".

TIP: The order in which **Values** and/or **Tags** are listed is taken into account during this process.

TIP: If a Sequence item (SQ) is used as one of the **Tags** to be searched, every child item in the sequence will be searched.

4.3.1 Defaults

The default values that are used for DICOM Parsing are reviewed in this section.

You can always revert to the defaults either by clicking the **Reset All** button on each property's individual configuration dialog, or by clicking the global **Reset** button in the **DICOM Parsing** section of the **Options** dialog.

4.3.1.1 Image Type

Tag(s) Searched

0008,103E	Series Description
0008, 1030	Study Description
0054, 0400	Image ID
0054, 0220	View Code Sequence
0008, 0008	Image Type

Searching for Value(s)

Transmission	-TR ATT MAP
Transaxial	TRANSVERSE
Short Axis	RECON GATED TOMO RECON TOMO SHORT AXIS
Planar Projection	GATED TOMO TOMO
Secondary Capture	SECONDARY

4.3.1.2 Acquisition Type

Tag(s) Searched

0054, 0016	Radiopharmaceutical Information Sequence
0040, 0555	Acquisition Context Sequence
0054, 0400	Image ID
0018, 1030	Protocol Name
0008, 103E	Series Description
0020, 0010	Study ID

Searching for Value(s)

AdreView	MIBG
Stress	STRESS EXERCISE GSPECT SGATE STR STR_
Rest	REST RST -R DLY REINJ RST_
FDG	FDG C-B1031 Fluorodeoxyglucose F ¹⁸
Redistribution	REDIST

4.3.1.3 Orientation

Tag(s) Searched

0054, 0410	Patient Orientation Sequence Code
0054, 0412	Patient Orientation Modifier Code Sequence
0020, 0020	Patient Orientation
0018, 1030	Protocol Name
0008, 103E	Series Description
0008, 1030	Study Description

Searching for Value(s)

Prone	-PRONE PRONE
Supine	SUPINE

4.3.1.4 Isotope

Tag(s) Searched

0018, 0031	Radiopharmaceutical
0054, 0016	Radiopharmaceutical Information Sequence
0054, 0012	Energy Window Information Sequence
0008, 103E	Series Description
0008, 1030	Study Description
0032, 4000	Study Comments

Searching for Value(s)

Tc99m	TC-99M TC99M MIBI 99MTC 99M
Tl201	TL-201 TL201 TL 201 TL 201
F18	F18 F-18 F 18 FLUORINE FDG
Rb82	RB RB82 RB-82 RB 82 RB^82 Rb-892
N13	N13 N-13 N 13
I123	123 IODINE I-123

4.3.1.5 Radiopharmaceutical

Tag(s) Searched

0018, 0031	Radiopharmaceutical
0054, 0016	Radiopharmaceutical Information Sequence
0054, 0012	Energy Window Information Sequence
0008, 103E	Series Description
0008, 1030	Study Description
0032, 4000	Study Comments

Searching for Value(s)

Tetrofosmin	TETROFOSMIN TETRO
Sestamibi	MIBI
MIBG	MIBG

4.3.1.6 AC (Attenuation Corrected)

Tag(s) Searched

0018, 1030	Protocol Name
0008, 103E	Series Description
0008, 1030	Study Description
0054, 0400	Image ID
0020, 0010	Study ID

Searching for Value(s)

AC	AC
	-AC
	IRAC

4.3.1.7 MC (Motion Corrected)

Tag(s) Searched

0018, 1030	Protocol Name
0008, 103E	Series Description
0008, 1030	Study Description
0054, 0400	Image ID
0020, 0010	Study ID
0020, 4000	Image Comments

Searching for Value(s)

MC	MC
	-MC
	MOCO
	MOTION

4.3.1.8 Dynamic

Tag(s) Searched

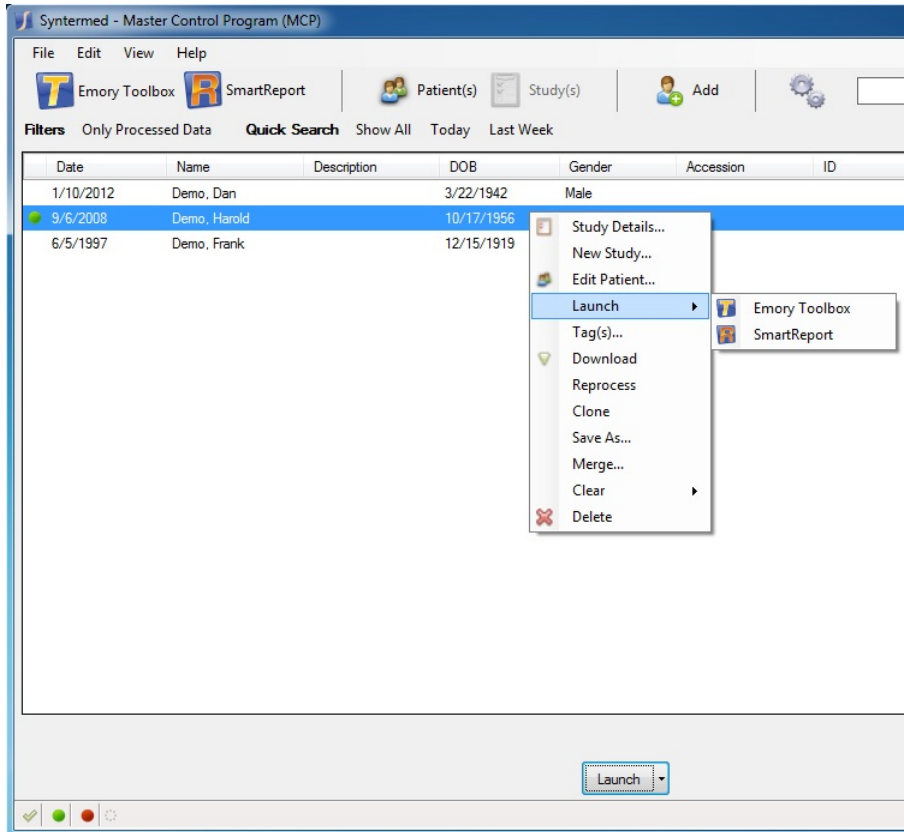
0018, 1030	Protocol Name
0008, 103E	Series Description
0008, 1030	Study Description
0054, 0400	Image ID
0020, 0010	Study ID

Searching for Value(s)

Dynamic	Dynamic
---------	---------

5 Launching Data

There are multiple ways to launch applications within MCP which are described below.



Double Click - each study is associated with an application so by double clicking on the patient the study will be launched with the appropriate application.



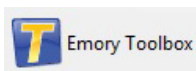
Selecting ICON - the study may also be selected and then click on the ICON located at the top of MCP:

Drop Down Window - the study can be launched by selecting the study then clicking right and then selecting Launch from the dropdown list as shown above.

Launch Button - Finally the study can be selected and then click on the Launch button at the bottom of the screen and select the application.

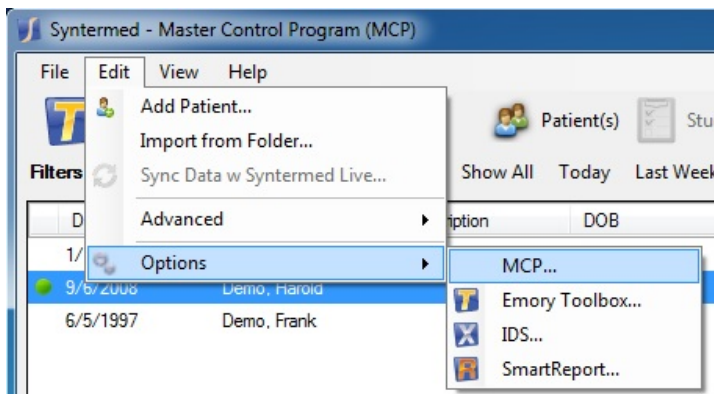
5.1 Plugins

The plugin applications are applications within MCP which patient data can be launched into. These applications are listed on the top of the MCP window.

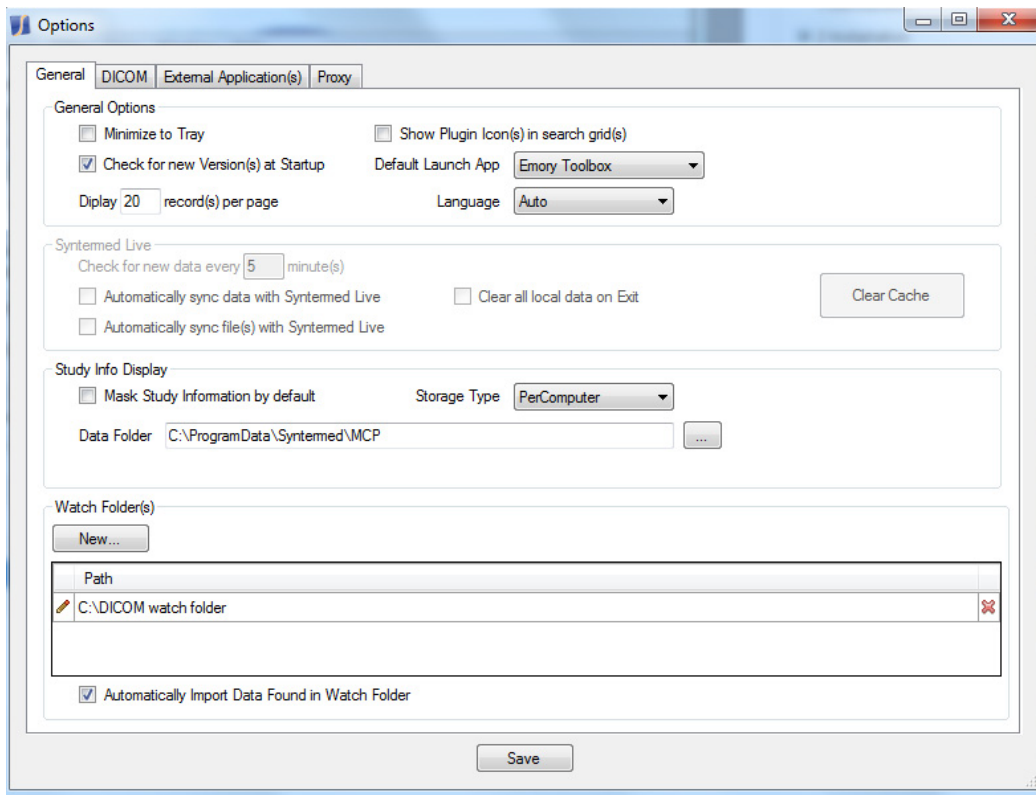


6 Options & Configuration

Each plugin may have its own set of configuration options which can be accessed from the Edit -> Options menu item in MCP. The list of application plugins is shown in this drop down list as shown below.

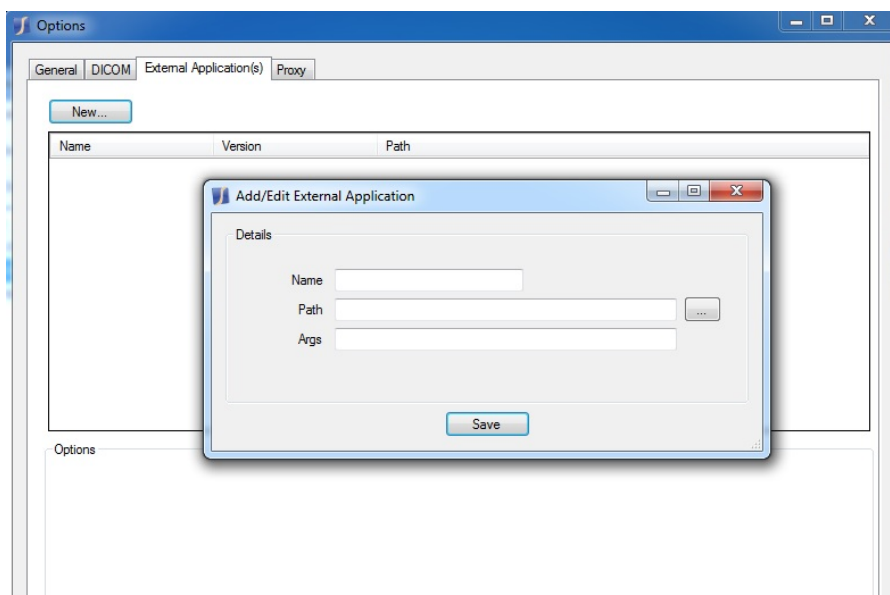


MCP has its own set of options which will be described in this section.



General - The general options screen allows for setting customized settings for MCP. There are General Options, Syntermed Live options if the user has Live privileges, Study Info Display options, and Watch Folder which is described in section 4.2.2.

DICOM - the DICOM options were described in section 4.2.1.



External Applications - This option allows the user to enter other applications that can be launched from MCP. Once the new application is entered and

saved the application can be selected by clicking on the Launch button at the bottom of the MCP window.

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